

B Faculty Dividing Powers

Time limit: 1s

Fred Faculty and Paul Power love big numbers. Day after day Fred chooses a random integer n and he computes $n!$. His friend Paul amuses himself by computing several powers of his randomly chosen integer k like k^2 , k^3 , ... and so on. On a hot summer day, Fred and Paul got really, really bored, so they decided to play a joke on their buddy Dave Divider. Fred chooses a random integer n while Paul chooses a random integer k . They want Dave to find the biggest integer i such that k^i divides $n!$ without a remainder, otherwise they will throw a cake in Dave's face. Because Dave does not like cakes in his face, he wants you to help him finding that integer i .

Input

The first line contains the number of test cases t ($1 \leq t \leq 100$). Each of the following t lines contains the two numbers n , k ($2 \leq n \leq 10^{18}$, $2 \leq k \leq 10^{12}$) separated by one space.

Output

For each test case, print the maximum integer i on a separate line.

Sample Input 1

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2
5 2
10 10
```

Sample Output 1

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3
2
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